

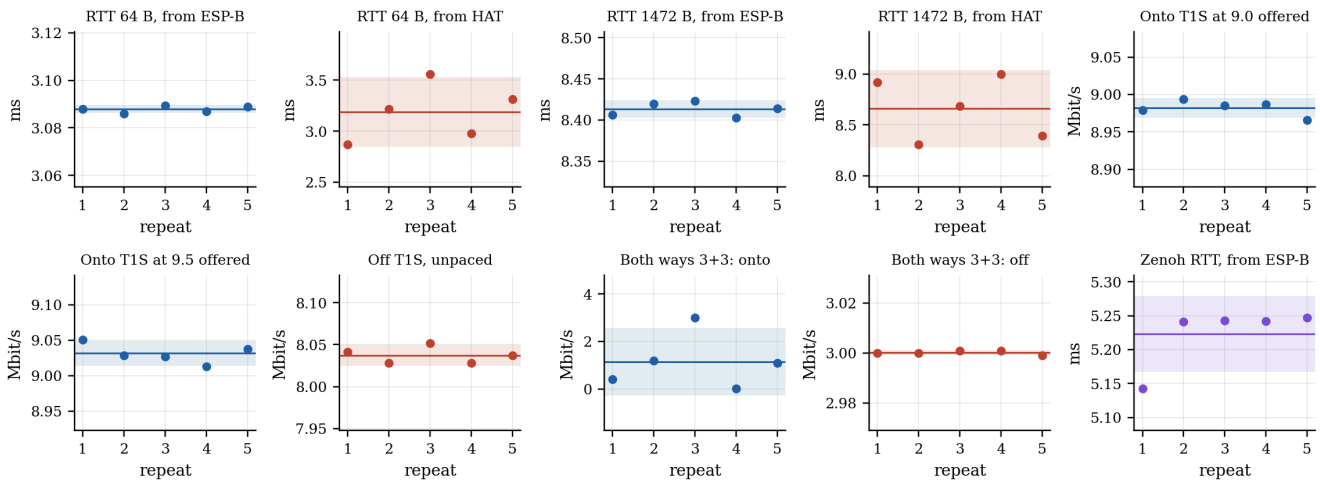
Two-board T1S campaign: repeatability, SPI model check, periodic traffic, tail, recovery

Run 2026-10-06 14:45:31 — campaign_20261006_144531.json, generated by make_campaign_report.py. Intervals are 95 % (Student t over the repeats).

Setup. esp32-5: ESP32-S3 + LAN8651 HAT, PLCA on (id 0 of 2), SPI 25 MHz asked, 20.00 MHz actual, 192.168.100.65. esp32-7: ESP32-S3 W5500 at 192.168.100.66 on the converter's 100BASE-TX port. Converter: LAN8670 + LAN9355, ID 1 / count 0 (not coordinating). All traffic generated and measured by the two boards; Zenoh's periodic traffic paused except in its own measurements.

1. Repeatability (5 runs, both boards rebooted between runs)

Fig. C1 Repeatability: 5 runs, both boards rebooted between runs; line = mean, band = 95 % CI



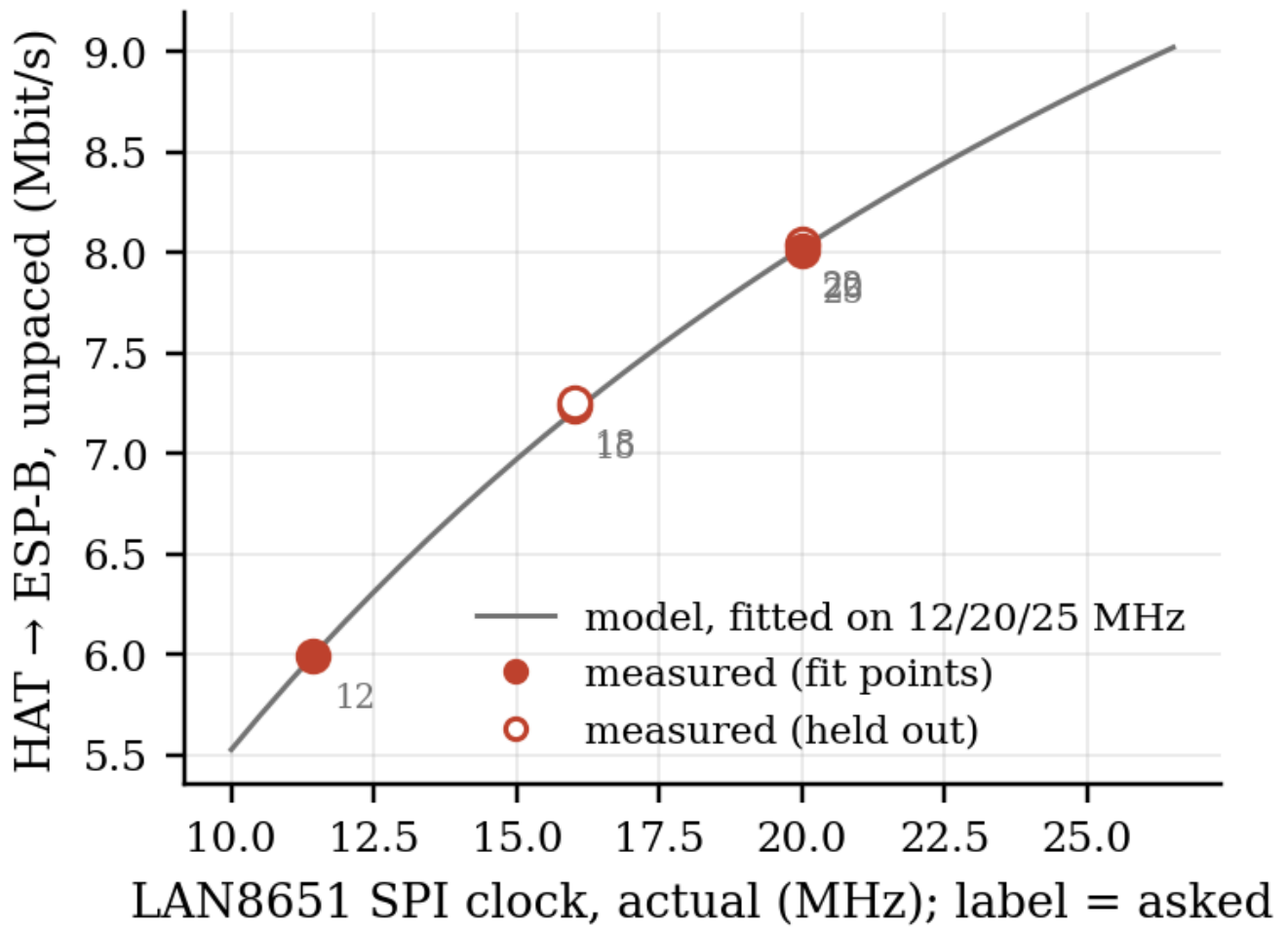
metric	per run	mean $\pm$ 95 % CI	spread (max–min)
RTT 64 B, from ESP-B	3.088 / 3.086 / 3.090 / 3.087 / 3.089	3.088 $\pm$ 0.002 ms	0.004
RTT 64 B, from HAT	2.867 / 3.219 / 3.558 / 2.975 / 3.314	3.187 $\pm$ 0.341 ms	0.691
RTT 1472 B, from ESP-B	8.406 / 8.420 / 8.423 / 8.402 / 8.414	8.413 $\pm$ 0.011 ms	0.021
RTT 1472 B, from HAT	8.915 / 8.303 / 8.685 / 8.998 / 8.392	8.659 $\pm$ 0.382 ms	0.694
Onto T1S at 9.0 offered	8.979 / 8.994 / 8.985 / 8.987 / 8.966	8.982 $\pm$ 0.013 Mbit/s	0.028
Onto T1S at 9.5 offered	9.051 / 9.029 / 9.027 / 9.013 / 9.038	9.032 $\pm$ 0.017 Mbit/s	0.038
Off T1S, unpaced	8.041 / 8.028 / 8.052 / 8.028 / 8.037	8.037 $\pm$ 0.012 Mbit/s	0.024
Both ways 3+3: onto	0.405 / 1.193 / 2.997 / 0.033 / 1.093	1.144 $\pm$ 1.418 Mbit/s	2.964
Both ways 3+3: off	3.000 / 3.000 / 3.001 / 3.001 / 2.999	3.000 $\pm$ 0.001 Mbit/s	0.002
Zenoh RTT, from ESP-B	5.143 / 5.241 / 5.243 / 5.242 / 5.247	5.223 $\pm$ 0.056 ms	0.104

Stable across reboots (95 % CI under 1 % of the mean): RTT 64 B, from ESP-B, RTT 1472 B, from ESP-B, Onto T1S at 9.0 offered, Onto T1S at 9.5 offered, Off T1S, unpaced, Both ways 3+3: off.

- **RTT 64 B, from HAT** moves between runs (2.87–3.56 ms) while staying tight within a run. The HAT's probe timer and ESP-B's 1 ms W5500 poll start at an arbitrary phase after each boot, and the reply waits for that poll: a reboot reshuffles the phase. Seen from ESP-B, which owns the poll, the same path is stable.
- **RTT 1472 B, from HAT** moves between runs (8.30–9.00 ms) while staying tight within a run. The HAT's probe timer and ESP-B's 1 ms W5500 poll start at an arbitrary phase after each boot, and the reply waits for that poll: a reboot reshuffles the phase. Seen from ESP-B, which owns the poll, the same path is stable.
- **Both ways 3+3: onto**: the converter's stream onto T1S fell short of the 3 Mbit/s offered in 4 of 5 runs (0.41 / 1.19 / 3.00 / 0.03 / 1.09 Mbit/s), from a total stall to none. The collapse is real but not deterministic; a single run can show anything from 0 to 3 Mbit/s, which is why repeats were needed to state it.
- **Zenoh RTT, from ESP-B**: 5.223 $\pm$ 0.056 ms.

2. SPI clock: model fitted on the 12/20/25 MHz settings, checked on 15/18/22 MHz

Fig. C2 SPI clock model, out-of-sample check



Model: every 1472 B datagram costs a fixed $t_0 = 828 \mu\text{s}$ plus $k/f = 13638 \mu\text{s} \cdot \text{MHz} / f$ of SPI time, fitted on the fit clocks only. With 24 TC6 chunks of 68 B per frame, k corresponds to $c = 1.04$ clock periods per transferred bit (1.00 would be the bare chunk transfer).

SPI MHz asked → actual	role	measured (Mbit/s)	model (Mbit/s)	error	512 B unpaced	onto T1S at 9.5	RTT 1472 B from HAT (ms)
12 → 11.43	fit	5.993	5.993	+0.0 %	4.899	6.599	9.85
15 → 16.00	held out	7.240	7.209	+0.4 %	5.714	8.045	8.46
18 → 16.00	held out	7.258	7.209	+0.7 %	5.705	8.056	8.50
20 → 20.00	fit	8.035	8.022	+0.2 %	6.226	9.036	8.39
22 → 20.00	held out	8.044	8.022	+0.3 %	6.240	9.033	8.42
25 → 20.00	fit	8.010	8.022	-0.2 %	6.244	9.033	8.42

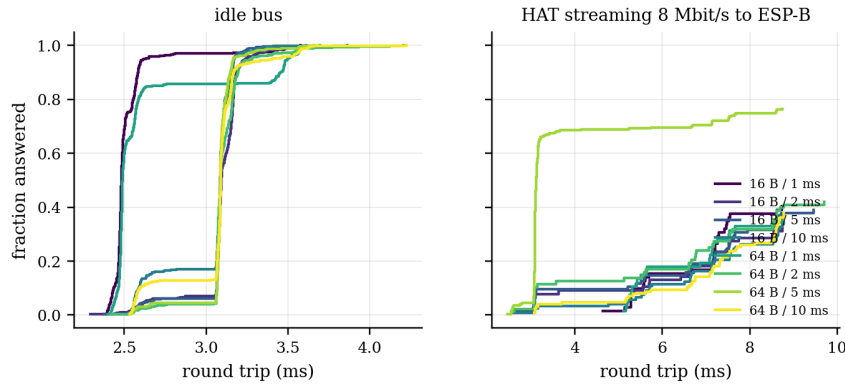
Held-out error: mean **0.5 %**, worst **0.7 %**.

The clock asked for is not the clock that runs. The ESP32-S3's SPI peripheral divides 80 MHz by an integer, and the firmware never lets it exceed the LAN8651's 25 MHz (DS60001734F, Table 9-9), so the boards ran: 12 → 11.43, 15 → 16.00, 18 → 16.00, 20 → 20.00, 22 → 20.00, 25 → 20.00. In spec the fit rests on two clocks (11.43

and 20 MHz) and the held-out settings add one new clock (16 MHz, asked as 15 and 18) and repeat 20 MHz (asked as 22): one prediction, one repeatability check. The model is fitted and drawn against the actual clock.

3. CAN-like periodic messages

Fig. C3 CAN-like periodic messages, round trip from ESP-B; loaded: only probes sent while the load ran (curves end below 1 where probes were lost)



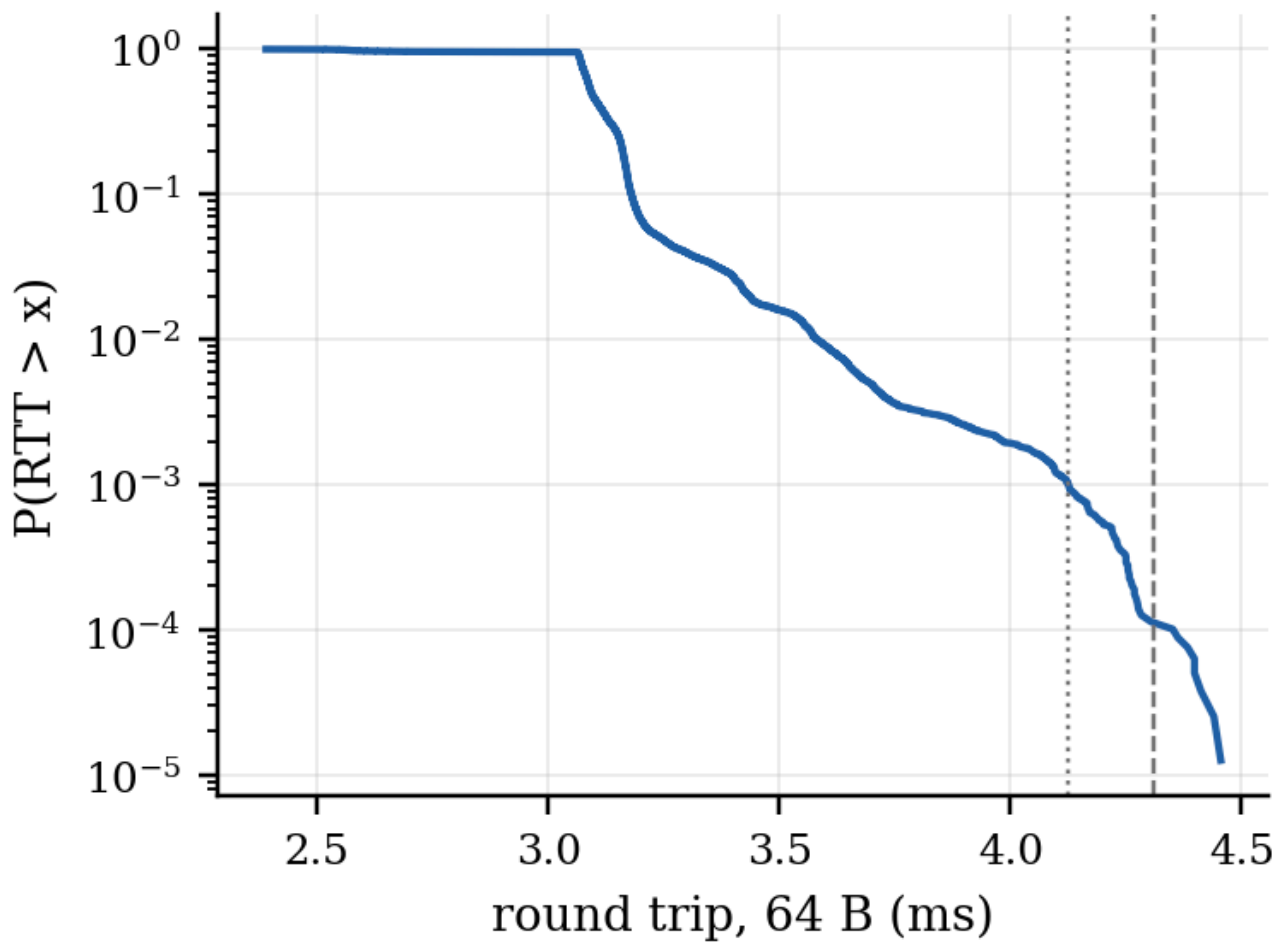
Sequential probes (one in flight), so the effective period is $\max(\text{period}, \text{round trip})$. Under load only the probes sent while the load was running are counted (each probe's start time is rebuilt from the intervals, answers and 100 ms timeouts). Deadline columns count a probe as missed if it was lost or its round trip exceeded the deadline. Bound: the idle median plus one worst-case wait behind a 1518 B frame in each direction (2×1.239 ms).

payload	period	load	probes	lost	median	p99	max	> idle + 2×wait	miss @ 5 ms	miss @ 10 ms
16 B	1 ms	idle	500	0	2.48	3.49	3.58	0	0.0 %	0.0 %
16 B	2 ms	idle	500	0	3.10	3.47	3.86	0	0.0 %	0.0 %
16 B	5 ms	idle	500	1	3.09	3.26	3.61	0	0.2 %	0.2 %
16 B	10 ms	idle	500	0	3.09	3.35	3.61	0	0.0 %	0.0 %
64 B	1 ms	idle	500	0	2.49	3.58	3.70	0	0.0 %	0.0 %
64 B	2 ms	idle	500	0	3.09	3.63	4.21	0	0.0 %	0.0 %
64 B	5 ms	idle	500	0	3.09	3.50	3.66	0	0.0 %	0.0 %
64 B	10 ms	idle	500	0	3.09	3.60	4.23	0	0.0 %	0.0 %
16 B	1 ms	8 Mbit/s	72	43	7.18	8.70	8.73	28	98.6 %	59.7 %
16 B	2 ms	8 Mbit/s	77	52	6.61	8.70	8.75	17	90.9 %	67.5 %
16 B	5 ms	8 Mbit/s	95	58	7.15	9.19	9.46	25	90.5 %	61.1 %
16 B	10 ms	8 Mbit/s	122	80	7.12	8.76	8.77	33	95.9 %	65.6 %
64 B	1 ms	8 Mbit/s	67	43	6.26	8.70	8.75	24	100.0 %	64.2 %
64 B	2 ms	8 Mbit/s	88	51	6.76	9.37	9.70	24	87.5 %	58.0 %
64 B	5 ms	8 Mbit/s	273	64	3.09	8.60	8.76	20	31.1 %	23.4 %
64 B	10 ms	8 Mbit/s	127	79	7.30	8.78	8.78	38	95.3 %	62.2 %

On an idle bus every probe returns, well inside a 5 ms deadline, and none exceeds the bound. Under an 8 Mbit/s stream from the node, **23–68 % of the probes sent during the load are lost**, and most of those that return still sit near the idle median. The losses are the converter's: each probe has to go onto T1S through it while the node streams the other way (§1, two-way). The answered probes that exceed the bound do so by queueing behind the node's own stream in its transmit path, not on the wire.

4. Tail (78,997 probes, 64 B every 2 ms, 15 min)

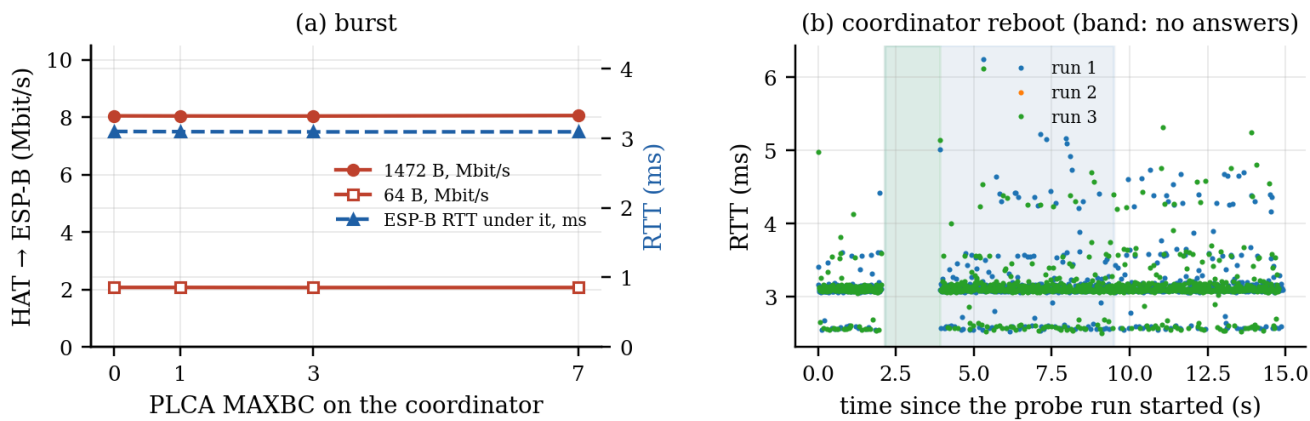
Fig. C4 Tail, 78,997 samples



median	p99	p99.9	p99.99	max	lost
3.094	3.582	4.125	4.309	4.455	3

5. PLCA burst and coordinator recovery

Fig. C5 PLCA burst and coordinator recovery



MAXBC	register read back	1472 B unpaced (Mbit/s)	64 B unpaced (Mbit/s)	ESP-B probes during the coordinator's 8 s flood: answered
0	0x000000080	8.043	2.069	73 of 139
1	0x000000180	8.039	2.069	41 of 110
3	0x000000380	8.037	2.064	49 of 117
7	0x000000780	8.055	2.066	55 of 123

While the coordinator floods unpaced, **only 37–52 % of the other node's probes are answered** at any MAXBC: they have to get onto T1S through the converter while the bus is busy in the other direction. They are all answered again once the flood ends.

Burst mode changes nothing measurable: with lwIP handing the MAC one frame per call over SPI, the coordinator rarely has a second frame ready inside the burst timer, so it never uses the extra slots.

run	probes	lost	longest run of lost probes	outage
1	1000	18	17	1.87 s (± 0.11)
2	—	—	—	no data (the prober's report was not captured)
3	1000	17	17	1.87 s (± 0.11)

Outage = the longest run of unanswered probes $\times$ (10 ms interval + 100 ms timeout), so $\pm$ one probe. Mean 1.87 s over 2 reboots with data (1 without).

The outage covers the coordinator's own reboot and bring-up (ESP32 boot, LAN8651 driver install, PLCA). Answers resume as soon as its first beacon goes out; the peer and the converter need no recovery action.

6. What two boards cannot show (stated, not hidden)

reviewer item	why not here	what it needs
more than 2 PLCA nodes (cycle vs N, fairness, empty slots)	one LAN8651 node exists	a second / third LAN8651 node (HAT Rev C)
converter removed (control experiment)	the only other T1S device is the converter	two LAN8651 nodes, or node + bridge pair
baseline without T1S	W5500 $\neq$ W5500 needs a crossover (no auto-MDIX, datasheet)	a crossover cable or a switch
cable length / stubs / termination	one 1 m cable	cables and a multidrop harness
TO_TIMER sweep	the converter's TO cannot be set; mismatched TO would measure the mismatch	two settable nodes
collision / error counters	the guessed MAC counter map was wrong and stalled TX	the LAN8651 statistics register map
CPU load and power	not instrumented	FreeRTOS run-time stats build, a power meter
Zenoh small-message rate (~1000 msg/s unbatched)	found (2026-10-06): the publisher's per-datagram put (~1 ms), not the subscriber; batching gives ~5500 msg/s (see the suite report)	latency of batched puts

- TO_TIMER not swept: the converter's TO cannot be set (its UART is output-only); a TO differing between nodes desynchronises PLCA, which would measure the mismatch, not TO.
- setup.node.spi_actual and spi_actual were added after the run from this run's own spi_sweep readings (the server ran campaign.py from before it recorded them); firmware 15cee5b clamps SCLK to ≤ 25 MHz.

Measurement corrections in this run: the sink's rate is now (all datagrams after the first) / (first-to-last arrival, μ s), removing the ~0.5 % over-read; arrival-gap histogram extended to 50 ms (20 μ s bins to 2 ms, 1 ms bins above).